

SYSTEM PROPOSAL

[Local Directory - Lost & Found : A Web-Based Lost and Found Directory for Icct Cainta]

Platform: Web Application

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1. Introduction

Students and staff may lose personal belongings such as IDs, school materials, gadgets, and other items while inside the campus. When lost and found information is posted only through Facebook Groups or messages, it can be difficult to find older posts and identify the current status of an item.

Local Directory - Lost & Found provides one website where users can view lost and found listings, search by item details, upload photos, and check the reported location. A map can help users understand where an item was lost or found, while an upload service can store item photos.

The proposed Web application will organize these records into a searchable local directory and provide basic item status management and administrator control.

2. Purpose of the Project

This proposal aims to present a Web application that will address the difficulty of finding and organizing lost and found item information at ICCT Cainta, streamline the process of posting and searching item reports, and improve communication between students and the campus community.

3. Objectives

3.1 General Objective

To design and develop a web-based lost and found directory that organizes item reports, supports searching and location information, and helps users identify and follow up on reported items.

3.2 Specific Objectives

- To provide users with registration and login for authenticated access.

- To allow users to create lost or found item reports with descriptions, categories, dates, locations, and photos.
- To provide search and filtering tools for finding relevant lost and found reports.
- To integrate a Maps API to display or identify reported item locations.
- To integrate an Upload API for storing photos attached to item reports.
- To allow users to update the status of their own item reports when an item is found, returned, or no longer active.
- To allow administrators to manage users and review or remove inappropriate item reports.

4. Project Scope

4.1 In Scope

- User registration and login.
- Lost and found item posting with item details, category, date, location, and status.
- Photo upload for item reports.
- Search and filtering of lost and found listings.
- Map-based location information using a Maps API.
- Item status updates such as active, found, returned, or closed.
- Administrator management of users and item reports.
- MySQL storage for user accounts and lost and found records.

4.2 Out of Scope

- Official guarantee that a reported item will be recovered.
- Payment, selling, or purchasing of found items.
- Automatic identification of item owners.
- Integration with private campus security or property-management systems.
- Real-time GPS tracking of users or lost items.
- A replacement for the official ICCT lost and found procedure.

5. System Features / Functions

Module	Feature	Description
User Management	Registration & Login	Allows users to create accounts and access the lost and found directory.
Item Reports	Lost & Found Posting	Allows users to report lost or found items with basic item information.
Item Details	Category & Status	Organizes reports by item category and current status.
Photo Upload	Item Photo Upload	Allows users to attach photos to lost and found reports through the Upload API.
Search	Search & Filters	Helps users find reports by item name, category, date, location, or status.

Module	Feature	Description
Maps	Location Map	Uses the Maps API to show or identify the reported location of an item.
Status	Report Status Upload	Allows users to update their reports when an item is found, returned, or closed.
Administration	User & Report Management	Allows administrators to manage users and review or remove reports.

6. Technology Stack

Layer	Technology / Tool	Purpose
Frontend (Web)	React, Vite, React Router	Builds the web interface for listings, reports, search, and user access.
Backend / API	Node.js , Express	Handles authentication, item reports, status updates, and API requests.
Database	MySQL	Stores users, item reports, categories, statuses, and related records.
External API	Maps API	Provides map and location information for reported items.
External API	Upload API	Handles photo uploads for lost and found item reports.
Cloud / Hosting	Render / Vertical (proposed)	Deployment options for the frontend and backend.
Version Control	Git / Github (proposed)	Stores and tracks source-code changes.
Project Management Tool	Trello / Notion (proposed)	Task tracking and project progress.

7. Target Users

User Role	Description	Key Needs / Usage
Student / Staff User	Posts, searches, and updates lost or found item reports.	Report items, search listings, view locations, and follow item status.
Admin	Manages users and oversees the lost and found directory.	Review reports, remove inappropriate posts, and manage user access.

8. Project Timeline / Work Plan

Phase	Activities	Estimated Duration
Planning & Requirements Gathering	Review the project workflow, users, requirements, and main features.	1 week
System Design	Plan the database, interface, user roles, API use, and system flow.	2 weeks
Development	Build and refine the frontend, backend, database, API integration, and core features.	4 weeks
Testing & QA	Test the main functions, calculations, API communication, and user permission.	2 weeks
Deployment	Configure the database. environment variables, hosting, and project documentation.	1 week
Project Presentation	Presentation of project	Ongoing / Final
